CSCE 46103/56103 Artificial Intelligence – Fall 2026

Supervised Learning

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This Material's Topics

- Labeled examples, hypotheses, training, and prediction.
- Comparing classification rules using observed errors.
- The limits of training data and input representations.
- Regression, squared loss, and robot demonstrations.

The Central Question

How can examples guide the choice of a prediction rule for future inputs?

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

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Labeled Examples and Hypotheses

- Input: $x \in \mathcal{X}$.
- Target: $y \in \mathcal{Y}$.
- Example: a pair (x, y).
- Training data: $D = \{(x_i, y_i)\}_{i=1}^{n}$.
- Hypothesis: $h : \mathcal{X} \to \mathcal{Y}$.
- Hypothesis space: a set $\mathcal{H}$ of allowed functions.

$$\hat{y} = h(x).$$

A Prediction and Its Target

$\hat{y}$ is the model's output. The observed target y is the value against which that prediction is compared.

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

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Classification and Regression

	Classification	**Regression**
Output	A class from a finite set	A numerical quantity
Restaurant example	Will the customer wait?	How many minutes until seating?
Robot example	Is the gripper open or closed?	What joint angle comes next?
Possible target	$y \in \{\text{Yes}, \text{No}\}$	$y \in \mathbb{R}$

Different Targets in One Setting

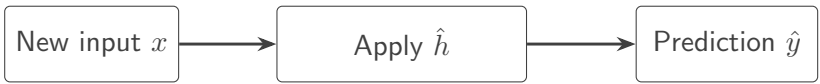
The prediction target determines which learning problem we formulate.

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

Training and Prediction

Training: choose a hypothesis using labeled examples





Prediction: use the selected hypothesis on an input

The new target is not available when making the prediction.

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

Six Restaurant Examples

Use two input attributes: $x = (\text{Patrons}, \text{WaitEstimate})$.

Example	Patrons	Wait estimate (min)	WillWait
x_1	Some	0–10	Yes
x_2	Full	30–60	No
x_3	Some	0–10	Yes
x_4	Full	10–30	Yes
x_5	Full	> 60	No
x_6	Some	0–10	Yes

The Target's Meaning

WillWait records the observed choice, not the optimal decision or actual waiting time.

Teaching subset of the restaurant data in Russell and Norvig [2020, Fig. 19.2]; two of the original attributes retained.

Three Candidate Hypotheses

Hypothesis	Prediction rule
h_0	Always predict Yes.
h_1	Predict Yes when Patrons is Some; otherwise predict No.
h_2	Predict Yes when WaitEstimate is in the category 0–10 or 10–30; otherwise predict No.

- For this exercise, $\mathcal{H} = \{h_0, h_1, h_2\}$.
- Each rule produces a prediction for an input.
- The category labels are used as given; we do not reinterpret their boundaries.

Teaching subset of the restaurant data in Russell and Norvig [2020, Fig. 19.2]; two of the original attributes retained.

Classification Loss

Zero–one loss counts an incorrect prediction as one mistake:

$$L(\hat{y}, y) = \begin{cases} 0, & \hat{y} = y, \\ 1, & \hat{y} \neq y. \end{cases}$$

The empirical risk is the average loss on the training examples:

$$\widehat{R}_D(h) = \frac{1}{n} \sum_{i=1}^{n} L(h(x_i), y_i).$$

Loss functions: Russell and Norvig [2020, Sec. 19.4].

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For Zero–One Loss

Empirical risk is the fraction of training examples classified incorrectly.

Loss functions: Russell and Norvig [2020, Sec. 19.4].

Comparing the Candidate Predictions

Example	Observed y	$h_0(x)$	$h_1(x)$	$h_2(x)$
x_1	Yes	Yes	Yes	Yes
x_2	No	Yes	No	No
x_3	Yes	Yes	Yes	Yes
x_4	Yes	Yes	No	Yes
x_5	No	Yes	No	No
x_6	Yes	Yes	Yes	Yes
Mistakes		2	1	0
$\widehat{R}_D$		$2/6$	$1/6$	$0/6$

A Better Fit to These Six Examples

The rule h_2 makes no errors on these six training examples.

Teaching subset of the restaurant data in Russell and Norvig [2020, Fig. 19.2]; two of the original attributes retained.

Empirical Risk Minimization

A learning objective can be written as

$$\hat{h} \in \arg\min_{h \in \mathcal{H}} \widehat{R}_D(h).$$

1. Specify the hypothesis space $\mathcal{H}$.
2. Specify the loss used to compare predictions and targets.
3. Use the training data to select a hypothesis with low empirical risk.

Loss functions: Russell and Norvig [2020, Sec. 19.4].

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Our Small Search

Among $\{h_0, h_1, h_2\}$, the minimizer is $\hat{h} = h_2$. Larger hypothesis spaces call for efficient search or optimization procedures.

Loss functions: Russell and Norvig [2020, Sec. 19.4].

Checkpoint: A New Restaurant Case

An additional case, x_{10}, has

$$x_{10} = (\text{Full}, \text{10–30}).$$

1. What does the selected rule h_2 predict?
2. Does its zero training error tell us the person's actual choice?

Additional case x_{10} from Russell and Norvig [2020, Fig. 19.2], with the same two retained attributes.

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Prediction and Observation

$h_2(x_{10}) =$ Yes, but the recorded label is No.

A perfect fit to the six training cases did not determine this new label.

This one case illustrates a limitation; it is not a generalization estimate.

Additional case x_{10} from Russell and Norvig [2020, Fig. 19.2], with the same two retained attributes.

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Information Lost in the Representation

Example	Patrons	Wait estimate (min)	WillWait
x_4	Full	10–30	Yes
x_{10}	Full	10–30	No

- The two cases have the same retained input and different labels.
- Every deterministic function of these two attributes must give them the same prediction.
- The original data contain additional attributes; our teaching representation discarded them.

A Representation Limit

More computation cannot recover distinctions missing from the input. This pair alone does not establish overfitting.

Cases x_4 and x_{10} from Russell and Norvig [2020, Fig. 19.2].

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Original Restaurant Data

Example	Alt	Bar	Fri	Hun	Patrons	Price	Rain	Res	Type	Wait est.	WillWait
x_1	Yes	No	No	Yes	Some	$$$	No	Yes	French	0–10	Yes
x_2	Yes	No	No	Yes	Full	$	No	No	Thai	30–60	No
x_3	No	Yes	No	No	Some	$	No	No	Burger	0–10	Yes
x_4	Yes	No	Yes	Yes	Full	$	Yes	No	Thai	10–30	Yes
x_5	Yes	No	Yes	No	Full	$$$	No	Yes	French	> 60	No
x_6	No	Yes	No	Yes	Some	$$	Yes	Yes	Italian	0–10	Yes
x_7	No	Yes	No	No	None	$	Yes	No	Burger	0–10	No
x_8	No	No	No	Yes	Some	$$	Yes	Yes	Thai	0–10	Yes
x_9	No	Yes	Yes	No	Full	$	Yes	No	Burger	> 60	No
x_{10}	Yes	Yes	Yes	Yes	Full	$$$	No	Yes	Italian	10–30	No
x_{11}	No	No	No	No	None	$	No	No	Thai	0–10	No
x_{12}	Yes	Yes	Yes	Yes	Full	$	No	No	Burger	30–60	Yes

Retained: Patrons and WaitEstimate (minutes). Orange: differences between x_4 and x_{10}.

Discarded: Alternate, Bar, Fri/Sat, Hungry, Price, Raining, Reservation, Type.

Restaurant data from Russell and Norvig [2020, Fig. 19.2, p. 657].

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Inductive Bias

Finite examples usually leave many possible prediction rules.

- A hypothesis space limits which functions can be selected.
- A learning procedure can prefer some functions over others.
- These restrictions and preferences form an inductive bias.

Examples of Restrictions

Use a linear function.
Use a decision tree of limited depth.

Examples of Preferences

Prefer a simpler rule when fits tie.
Expect similar inputs to have similar outputs.

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

Training Error and Future Error

Observed Training Examples

$$\widehat{R}_D(h) = \frac{1}{n}\sum_{i=1}^{n} L(h(x_i), y_i).$$

We can calculate this value from D.

Future Examples

$$R(h) = \mathbb{E}_{(X,Y)\sim P}[L(h(X), Y)].$$

P describes the distribution of future inputs and targets.

- Generalization concerns predictions on examples beyond those used to learn.
- A small training error alone does not establish a small future error.
- Changes in the data source or environment can change future performance.

Loss functions: Russell and Norvig [2020, Sec. 19.4].

Underfitting and Overfitting

Underfitting

The learned rule fails to capture useful structure, even in the training examples.

- A model can be too restrictive.
- Training error can remain high.

Overfitting

The learned rule captures peculiarities of the training sample that do not generalize.

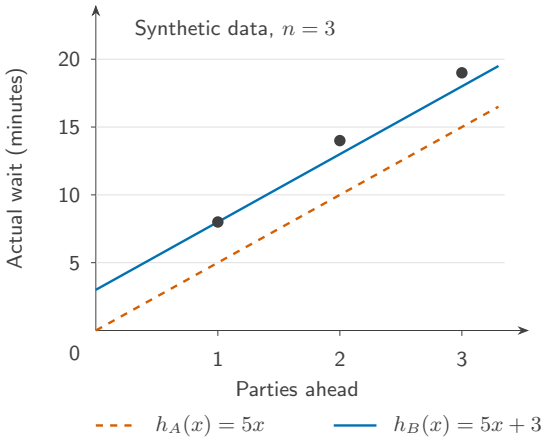
- A model can fit incidental variation.
- Training error can be misleadingly low.

Evidence Needed

Zero training error by itself proves neither good generalization nor overfitting. Evaluation on suitable separate data is required.

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

A Numerical Target



- Input x: number of parties ahead.
- Target y: actual minutes until seating.
- Two candidate linear functions:

$$h_A(x) = 5x, \qquad h_B(x) = 5x + 3.$$

Linear regression: Russell and Norvig [2020, Sec. 19.6.1]. Numerical data are synthetic teaching examples.

Squared Loss and Mean Squared Error

For a numerical target, one choice is squared loss:

$$L(\hat{y}, y) = (\hat{y} - y)^2.$$

Its average over the data is mean squared error (MSE):

$$\widehat{R}_D(h) = \frac{1}{n} \sum_{i=1}^{n} \big(h(x_i) - y_i\big)^2.$$

- $h(x_i) - y_i$ is the signed prediction error.
- Squaring prevents positive and negative errors from canceling.
- With targets in minutes, MSE has units of minutes squared.

Linear regression: Russell and Norvig [2020, Sec. 19.6.1]. Numerical data are synthetic teaching examples.

Comparing Regression Hypotheses

Parties ahead x	**Actual** y **(min)**	$h_A(x) = 5x$	$h_B(x) = 5x + 3$
1	8	5	8
2	14	10	13
3	19	15	18

Linear regression: Russell and Norvig [2020, Sec. 19.6.1]. Numerical data are synthetic teaching examples.

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3	19	15	18

$$\widehat{R}_D(h_A) = \frac{(-3)^2 + (-4)^2 + (-4)^2}{3} = \frac{41}{3} \approx 13.67 \text{ min}^2,$$

$$\widehat{R}_D(h_B) = \frac{0^2 + (-1)^2 + (-1)^2}{3} = \frac{2}{3} \approx 0.67 \text{ min}^2.$$

h_B has lower empirical squared loss on these three examples.

Linear regression: Russell and Norvig [2020, Sec. 19.6.1]. Numerical data are synthetic teaching examples.

Checkpoint: The Scale of an Error

The actual wait is 20 minutes. Compare two predictions:

Prediction	Absolute error	Squared loss
15 min	5 min	$5^2 = 25$ min^2
10 min	10 min	$10^2 = 100$ min^2

How does doubling the error magnitude change the squared loss?

Loss functions: Russell and Norvig [2020, Sec. 19.4].

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A Fourfold Increase

$(2e)^2 = 4e^2$. Squared loss gives large errors disproportionately greater weight than small errors.

Loss functions: Russell and Norvig [2020, Sec. 19.4].

Loss Functions and Task Goals

Loss	What it measures	What to consider
Zero–one	Frequency of incorrect predictions	It treats all mistakes equally.
Squared	Magnitude of numerical errors	Large errors receive more weight.

- The loss defines what the learner is asked to improve.
- The application determines which errors matter and how much.
- A low prediction loss need not imply a good downstream decision.

Loss functions: Russell and Norvig [2020, Sec. 19.4].

Supervised Learning from Demonstrations

For a robot, a demonstration can provide an input–target pair:

$$x = (\text{camera image}, \text{joint state}, \text{instruction}), \qquad y = \text{expert action}.$$

- A discrete action label gives a classification problem.
- A joint-angle or motion target gives a regression problem.
- Training compares the model's action with the demonstrated action.

A supervised-learning formulation applied to embodied agents; see Russell and Norvig [2020, Secs. 19.1–19.2].

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Prediction Within an Interaction Loop

The robot's action changes its next observation. Small action errors can lead to situations absent from the demonstrations.

Matching demonstrations and completing the task are distinct measures.

A supervised-learning formulation applied to embodied agents; see Russell and Norvig [2020, Secs. 19.1–19.2].

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Checkpoint: Formulating Robot Learning

A robot learns to place a cup on a tray. Each recorded step contains a camera image, joint positions, an instruction, and the expert's next joint positions.

1. Identify x, y, and a possible loss.
2. What is missing if the camera image is removed from x?
3. Does a low training loss establish successful cup placement?

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3. Does a low training loss establish successful cup placement?

One Formulation

x: image, current joint positions, instruction.
y: expert's next joint positions; a possible loss is summed squared angle error.
Without the image, object locations may be unavailable. Task success still needs separate evaluation during interaction.

Main Takeaways

- Supervised learning uses labeled examples to select a hypothesis.
- The representation, hypothesis space, and loss define the learning problem.
- Empirical risk measures fit to the available training examples.
- Generalization concerns examples beyond those used to learn.
- Prediction quality and success in the application require careful comparison.

Supervised learning: Russell and Norvig [2020, Sec. 19.2].

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Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL `https://aima.cs.berkeley.edu/`.